

DESi ablation Phase-5 — LONG documents (~18k tok), large + small model

Real run · temp 0 · 2 reps · 5 lifecycle/supersession cases padded to ~18k tokens (the reversed decision + ruled-out item buried in ~990 distractor turns). Now the budget-match BINDS: A sees the full ~18k chat; B / retrieval get ~372 tokens, so retrieval can pack only 8–9 of ~990 chunks (~1%). Sonnet-4.5 + Granite-4.1-8B. Conservative.

Mean recall (5 cases) — and input tokens

model	A full-chat	B DESi	B_auto	R1 bm25	R2n neural	E	A tok	B tok
Sonnet-4.5 (large)	0.88	1.0	0.96	0.52	0.08	1.0	18342	372
Granite-4.1-8B (small)	0.76	0.96	0.88	0.44	0.1	0.94	18342	372

Key deltas (mean recall over 5 cases)

comparison	note	Sonnet-4.5	Granite-4.1-8B
B – A	DESi vs full chat (49× fewer tokens)	+0.12	+0.20
B – R1	vs BM25 at 1% budget	+0.48	+0.52
B – R2n	vs small-neural at 1% budget	+0.92	+0.86
B – B_auto	DESi vs AUTO-built DESi	+0.04	+0.08
A – R2n	full chat vs budget retrieval	+0.80	+0.66

Findings (long-document regime) — conservative

1. At long context, budget-matched retrieval COLLAPSES; DESi holds at ~2% of the tokens. B = 1.0 (Sonnet) / 0.96 (Granite) on ~372 input tokens; full chat A = 0.88 / 0.76 on ~18,342 tokens (49× more) and is actually WORSE than B. Budget-matched retrieval at 1% of the chunks: BM25 ≈ 0.44–0.52, small-neural (bge-small) ≈ 0.08–0.10. So DESi delivers better-than-full-context recall at ~1/49 the cost — and beats budget retrieval by ~0.5 (BM25) to ~0.9 (neural).

2. This resolves the k-question honestly. The earlier k-sweep showed retrieval can match DESi on SHORT chats (optimal k ≈ all chunks is cheap). On LONG documents that is impossible: 'all chunks' does not fit the budget, and at 1% the small embedder is fooled by generically-relevant filler (it retrieves 'Unrelated sidebar... how is X set up' over the actual decision turns; BM25's keyword match is more robust). DESi's advantage is therefore TOKEN EFFICIENCY + robustness, and it is LARGE precisely in the long-document regime DESi is meant for.

3. Auto-construction survives long context. B_auto = 0.96 (Sonnet) / 0.88 (Granite): one extraction pass still finds and resolves the supersession buried in ~990 turns. B–B_auto = +0.04 / +0.08 (within/near noise). So the live-state slice is auto-constructible even from long, noisy documents — and a small model (Granite) does it nearly as well.

4. Confidently-wrong tracks retrieval failure. confidence_while_wrong: R2n = 0.6 / 0.2, R1 = 0.4 / 0.6, while A = B = B_auto = 0. Retrieval confidently reports filler when it misses the needles; the resolved state removes that failure mode.

Caveats. The neural collapse is for a SMALL embedder (bge-small, 384d) with a generic-task query; a stronger embedder or query expansion would rank needles better — so R2n=0.08 is a weak-retrieval floor, and B–R1=+0.5 is the more representative gap. N=5, 2 reps → directional. B's slice is curated, but B_auto shows it is constructible. The filler is synthetic distractor.

Net across short+long: DESi's value over retrieval is (i) TOKEN EFFICIENCY — full-context-or-better recall at ~2% of the tokens — and (ii) ROBUSTNESS on long documents where budget retrieval cannot fit / find the needles; both are auto-constructible. It is NOT a large absolute-capability win at unlimited budget (short-doc k-sweep), and metadata governance stays unestablished. No 'smarter' / 'wins generally' claim. Cost: ~130 calls.